

Automated Detection and Characterization of Microplastics in Fluorescence Microscopy Images Using MATLAB-Based Image Analysis

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Abstract. Microplastic pollution has become a major environmental concern due to its persistence and widespread occurrence in aquatic ecosystems. Conventional identification methods such as FTIR and Raman spectroscopy provide accurate characterization but are often costly and time-consuming. This work presents the development of **PlastID**, an automated MATLAB-based framework for the detection, segmentation, characterization and classification of microplastic particles in fluorescence microscopy images. The proposed methodology combines image enhancement, hybrid thresholding combining Otsu and local adaptive thresholding via logical conjunction, morphological filtering, watershed-based aggregate separation and geometric feature extraction. A rule-based classification system was implemented to identify fibers, fragments, films, beads and aggregates automatically. A graphical user interface supporting both single-image and batch-processing workflows was developed, together with automatic report generation. Experimental results demonstrate the capability of the framework to process large image collections efficiently while producing quantitative datasets and statistical summaries that support environmental monitoring applications.

Keywords: Microplastics · Image Processing · Fluorescence Microscopy · MATLAB · Segmentation · Particle Classification · Environmental Monitoring · Batch Processing

1 Introduction

Plastic waste is one of the most persistent challenges in contemporary environmental management. As larger plastic objects fragment over time, they produce microplastic particles — defined as particles smaller than 5 mm [1] — that accumulate in aquatic systems with no effective natural degradation pathway. Unlike many pollutants, microplastics do not dissolve or degrade at ecologically relevant timescales, and their small size allows them to be ingested by organisms across virtually all trophic levels [14,5].

The scale of the problem is becoming clearer as detection methods improve. Microplastics have been identified in freshwater rivers, marine sediments, drinking water, and, more recently, in human blood [13]. Wastewater treatment plants

(WWTPs) represent a critical intervention point: they remove a substantial fraction of microplastics before discharge, but are far from fully effective, and the particles that pass through continue to reach aquatic environments [4,23].

The bottleneck is not treatment technology alone — it is measurement. Current gold-standard methods such as micro-FTIR and Raman spectroscopy can accurately identify polymer type and particle morphology, but they are slow, expensive, and require specialist operators [19,2]. Processing a single sample can take hours, which makes routine, large-scale monitoring at WWTPs practically unfeasible. Image-based approaches have therefore attracted growing interest as a faster and cheaper alternative [17], even if they sacrifice chemical identification in exchange for throughput.

This work sits within that context. Developed under the RESOLVE project at the Universidade do Minho, it presents **PlastID**, a MATLAB-based framework that automates the detection, segmentation, morphological characterization, and classification of microplastic particles in Nile Red fluorescence microscopy images. The system processes images through a classical image-processing pipeline and exports per-particle metrics, classification results, and statistical reports — both for individual images and for large batches — without requiring manual intervention or annotated training data.

1.1 Objectives

The main objectives of this work are:

- Develop an automated computational methodology for the detection and quantification of microplastics in fluorescence microscopy images;
- Extract quantitative morphological descriptors (area, length, width, sphericity, compactness, eccentricity) for each detected particle;
- Implement an automatic rule-based classification system capable of categorizing particles into fibers, fragments, films, beads and aggregates;
- Develop a graphical user interface (GUI) to facilitate interaction with the system by non-specialized users;
- Implement a batch-processing module enabling large-scale analysis of image datasets with automatic report generation;
- Validate the framework in the context of the RESOLVE project, supporting scalable monitoring in wastewater treatment environments.

2 State of the Art

The detection and monitoring of microplastics have evolved significantly, transitioning from traditional spectroscopic techniques to advanced methodologies based on image analysis and machine learning [20]. This section reviews the main approaches and identifies the limitations that motivated the present work.

2.1 Traditional Spectroscopic Techniques

Spectroscopic approaches, including micro-FTIR and Raman spectroscopy, remain the gold standard for microplastic identification due to their high chemical specificity and accurate polymer characterization capabilities [17,25]. These methods allow reliable differentiation between polymer types, which is essential for environmental impact assessment.

Despite their high accuracy, these techniques present significant limitations. High operational costs, low throughput, and reliance on expert knowledge hinder their large-scale application [17,2]. Additionally, sample preparation and analysis times make them unsuitable for real-time monitoring, especially in dynamic environments such as WWTPs [4,23].

2.2 AI-Assisted Microscopy

To overcome the limitations of spectroscopic techniques, classical image processing methods were initially explored. These approaches rely on manually defined features such as color, morphology, and texture to identify microplastic particles [20]. Although computationally efficient, they exhibit only moderate accuracy and low robustness in heterogeneous samples with variable illumination and high noise.

The introduction of deep learning, particularly Convolutional Neural Networks (CNNs), represented a significant advancement. These methods enable automatic feature extraction and improved generalization, achieving high detection accuracy under diverse imaging conditions [27,26,9]. Compared to classical approaches, they offer a better balance between accuracy and scalability, although they strongly depend on the availability of high-quality annotated datasets [12,6].

2.3 Object Detection and Segmentation

Within deep learning, two main approaches can be considered, each suited to different analytical needs. Object detection approaches such as YOLO prioritize real-time performance and computational efficiency [3]. These are particularly suitable for portable or embedded systems, enabling rapid sample analysis. However, they provide only moderate precision in the exact delineation of particles.

On the other hand, segmentation-based approaches, such as Mask R-CNN and U-Net, perform pixel-level classification, allowing precise delineation of microplastics [26,27]. They enable accurate quantification and detailed morphological analysis, albeit with higher computational cost and lower real-time capability. This highlights a clear trade-off between speed and precision.

2.4 Hybrid Approaches

Hybrid approaches combine different detection modalities to leverage their complementary advantages. For instance, integrating spectral imaging with deep

learning enables both morphological and chemical characterization of microplastics [11]. Similarly, the use of photoacoustic imaging with automated preprocessing has shown improvements in detection within complex samples [10]. These approaches are particularly advantageous when visual information alone is insufficient, such as in distinguishing visually similar polymers or detecting particles in complex matrices [11,10]. However, they introduce additional complexity, requiring careful calibration and specialized hardware.

2.5 Datasets and Data Augmentation

The performance of deep learning models strongly depends on the availability of representative and well-annotated datasets. Recent works have introduced microscopy datasets for microplastic analysis, contributing to benchmarking and validation [12]. Nevertheless, significant challenges remain, including small dataset sizes, limited diversity, and acquisition under controlled conditions [12]. These issues hinder generalization to real-world data. To mitigate these limitations, data augmentation techniques are employed, including geometric transformations and Generative Adversarial Networks (GANs) [6].

2.6 Synthesis and Comparative Evaluation

Table 1 summarizes the main approaches reviewed.

Table 1. Comparison of microplastic detection approaches.

Method	Accuracy	Automation	Scalability	Limitations
Micro-FTIR [17]	VH	L	L	Expensive, slow, expert-dependent
Raman Spectroscopy [25]	VH	L	L	Costly, sensitive to interference
Classical Image Processing	M	M	H	Low robustness in complex samples
CNN-based Detection [27,26]	H	H	MH	Requires labelled datasets
Hybrid (DL + Spectral) [11]	VH	M	M	Complex integration and validation
Real-time Detection (YOLO) [3]	M	H	H	Limited validation in diverse conditions

VH=Very High, H=High, M=Medium, L=Low, MH=Medium-High.

3 Research Gap

The survey above points to a recurring tension in the field: methods that are accurate enough for scientific purposes tend to be too slow and resource-intensive

for routine monitoring, while faster automated approaches have so far struggled to generalize beyond the conditions under which they were trained [12,4,23].

Deep learning models illustrate this problem well. Training a segmentation network requires hundreds to thousands of annotated microscopy images, yet publicly available microplastic datasets remain small and largely acquired under controlled laboratory conditions [12,15]. When these models are deployed on real WWTP samples — which contain organic debris, irregular illumination, and highly variable particle concentrations — performance frequently degrades in ways that are difficult to predict or diagnose.

A second issue is fragmentation. Many published systems address only one stage of the analysis pipeline: a segmentation model here, a classification module there. End-to-end systems that handle image acquisition, preprocessing, particle separation, morphological extraction, classification, and reporting within a single deployable tool remain rare [17]. This fragmentation creates practical barriers for laboratories that lack deep learning expertise.

Hybrid approaches that combine image analysis with spectroscopic data are promising but introduce their own constraints: they require additional hardware, careful cross-modal calibration, and substantially more computational resources [11,10]. For most WWTP monitoring contexts, this level of complexity is neither necessary nor affordable.

The approach proposed here deliberately avoids these pitfalls. By relying on classical image processing rather than deep learning, the system requires no training data, runs on standard hardware, and produces deterministic, interpretable results. The trade-off — the loss of polymer-specific chemical identification — is acceptable for the intended use case: rapid, high-throughput morphological screening as a complement to, rather than a replacement for, spectroscopic analysis.

4 Materials and Methods

4.1 Dataset and Image Acquisition

The dataset consists of fluorescence microscopy images of microplastic particles stained with Nile Red, a fluorescent dye with high affinity for hydrophobic polymer surfaces that is widely used for fluorescence-based microplastic detection [16,7]. Images were acquired using a fluorescence microscope, with particle visibility enhanced against the dark background by the staining procedure. Figure 1 shows a representative image from the dataset.

Images were acquired in TIFF format, which embeds spatial calibration metadata (pixels per unit and physical unit, e.g. $\mu\text{m}/\text{pixel}$). **PlastID** automatically extracts this calibration factor from the image metadata, enabling all morphological measurements to be expressed directly in physical units (μm and μm^2) without requiring manual scale input from the user.

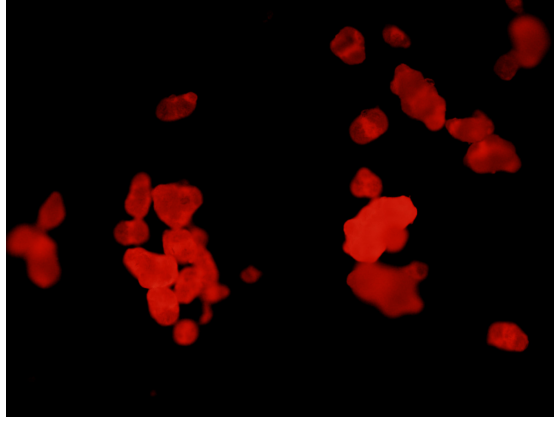


Fig. 1. Representative fluorescence microscopy image of Nile Red stained microplastic particles used in this work.

4.2 Image Preprocessing

The preprocessing pipeline begins with RGB channel decomposition and fluorescence channel selection. Since Nile Red staining primarily emits in the red-orange spectrum when bound to hydrophobic polymer surfaces [16], the working channel is computed as:

$$I_{\text{ch}} = I_R - 0.7 \cdot I_G \quad (1)$$

which suppresses green background autofluorescence while preserving the red fluorescence signal characteristic of stained microplastic particles. A 120×120 pixel region at the lower-right corner — which consistently contains microscope scale bar markings — is masked to zero before further processing.

Noise reduction is optionally performed using Wiener adaptive filtering [8], which suppresses additive noise while preserving edge sharpness better than uniform smoothing filters. Contrast enhancement subsequently applies morphological Top-Hat and Bottom-Hat filtering with a disk-shaped structuring element of radius 10 pixels [22], followed by histogram adjustment and Contrast Limited Adaptive Histogram Equalization (CLAHE) with a clip limit of 0.02 and a tile grid of 8×8 blocks [28], which compensates for uneven illumination fields common in fluorescence microscopy. Figure 2 illustrates the effect of the enhancement pipeline on a representative image.

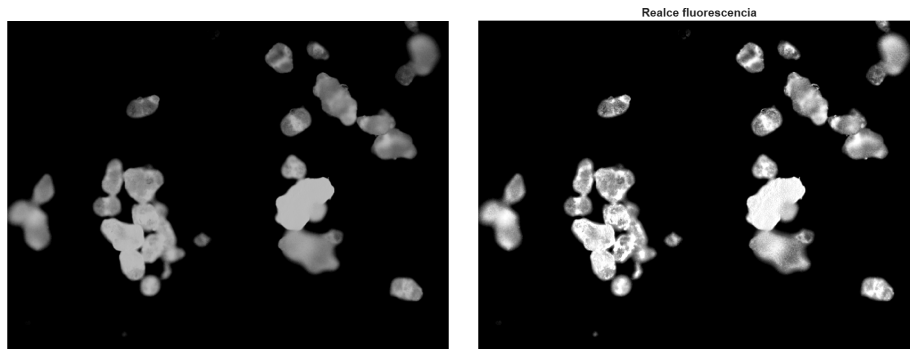


Fig. 2. Effect of the fluorescence enhancement pipeline. Left: raw fluorescence channel after colour extraction. Right: after Top-Hat/Bottom-Hat filtering and CLAHE.

4.3 Hybrid Thresholding

Microplastic particles were segmented using a hybrid strategy implemented in the `bin_itr` module, combining a brightness-adapted Otsu global threshold with a local adaptive threshold. The Otsu threshold T_{Otsu} is computed via `graythresh` and modulated by a brightness factor:

$$f = 0.80 + 0.10 \cdot \bar{I} \quad (2)$$

where $\bar{I}$ is the mean image intensity after preprocessing, yielding an effective global threshold $T_{\text{eff}} = T_{\text{Otsu}} \cdot f$. A local adaptive threshold T_{adapt} is independently computed using `adaptthresh` with sensitivity 0.70. The final binary mask is obtained by logical conjunction:

$$B_{\text{final}} = B_{\text{Otsu}} \wedge B_{\text{adapt}} \quad (3)$$

This strategy keeps only regions that both methods independently classify as foreground, suppressing spurious detections while retaining genuinely fluorescent particles [21,18]. Small isolated regions below 30 pixels are removed with `bwareaopen`, and a final erosion with a disk of radius 1 pixel refines particle boundaries.

Figure 3 illustrates the three intermediate binary masks produced by this pipeline for a representative image.

4.4 Morphological Refinement

The binary mask is further refined through morphological reconstruction using an eroded seed image, followed by an opening operation with a disk of radius 1, and hole filling [22]. Particles with area below 40 pixels are discarded.

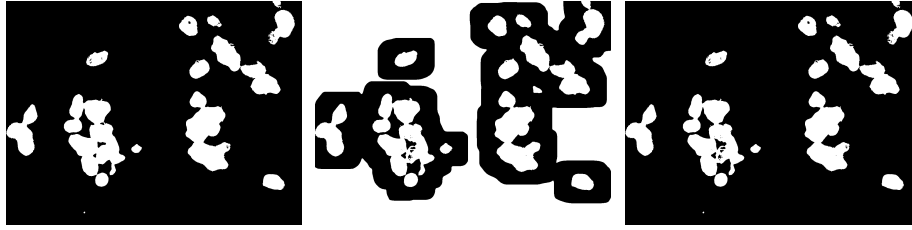


Fig. 3. Binary masks produced by the hybrid thresholding pipeline. Left: Otsu global mask. Centre: local adaptive mask. Right: final mask after logical conjunction.

4.5 Aggregate Separation

To address particle clustering, a selective watershed strategy is applied only to objects whose area exceeds twice the mean particle area of the current image. For each such aggregate, the distance transform is computed, smoothed with a Gaussian filter ($\sigma = 2$), and regional maxima above 20 pixels are used as markers for watershed segmentation [24]. Restricting watershed to oversized objects prevents over-segmentation of isolated particles.

4.6 Morphological Feature Extraction

Prior to feature extraction, spatial calibration is automatically retrieved from the TIFF image metadata. The scale factor ($\mu\text{m}/\text{pixel}$) is applied to all pixel-based measurements, converting area values to μm^2 and length measurements (major axis, minor axis, perimeter) to μm . This ensures that reported morphological descriptors are physically meaningful and comparable across images acquired at different magnifications.

For each detected particle, the following geometric descriptors are extracted via `regionprops`: Area, Major Axis Length, Minor Axis Length, Perimeter, Solidity, and Eccentricity. Additional descriptors computed from these include:

- **Circularity:** $C = 4\pi A / (P^2 + \varepsilon)$
- **Sphericity** and **Compactness:** derived via the `euclidean` module from convex hull geometry
- **Aspect Ratio:** Major Axis / Minor Axis

A colour criterion is additionally applied: for red-channel fluorescence, a particle is accepted only if the mean red channel intensity within its mask satisfies $\bar{I}_R / (\bar{I}_G + \varepsilon) > 1.15$, filtering out non-fluorescent debris.

4.7 Automatic Classification

Detected particles are classified into five categories using a rule-based decision tree applied in strict priority order:

1. **Aggregate:** $N_{\text{maxima}} \geq 3$ and Solidity < 0.80

2. **Fiber:** Eccentricity > 0.95 and Aspect Ratio > 3
3. **Bead:** Sphericity > 0.85 , Compactness > 0.75 , Solidity > 0.85 and Circularity > 0.75
4. **Film:** Compactness < 0.45
5. **Fragment:** all remaining particles

Aggregate detection is based on the number of regional maxima N_{maxima} in the distance transform of each object, which indicates the presence of multiple fused particles within a single connected component.

4.8 Graphical User Interface and Analysis Modes

PlastID was developed using MATLAB App Designer, providing a graphical user interface that allows users to select the fluorescence channel (*red*, *green*, or *orange*), configure preprocessing options (Wiener filtering on/off), and choose between two analysis modes.

In **single-image mode** (`opcao.avanco = 'imagem'`), the user selects an individual file through an interactive dialog (`uigetfile`, supporting `.bmp`, `.tif` and `.jpg`). After processing, segmentation results are displayed in real time as MATLAB figures, including the binary particle mask, labelled individual objects, and distribution histograms for area and particle length. Per-particle morphological metrics are reported in the Command Window, and results are saved to disk via `gravar` and `gravar_param`. An individual HTML report is also automatically generated for the analysed image.

In **batch-processing mode** (`opcao.avanco = 'series'`), the system iterates automatically over all images in a user-selected folder without further interaction. A global Excel dataset (`Resultados_Globais.xlsx`), global distribution charts, and a comprehensive HTML report (`relatorio_global.html`) with per-class statistics and percentages are generated at the end of the run.

This dual-mode design makes the framework suitable both for detailed inspection of individual samples and for high-throughput environmental monitoring campaigns.

5 Results and Discussion

5.1 Detection and Segmentation Performance

The hybrid thresholding strategy proved to be the most consequential design decision in the pipeline. A global threshold alone systematically misses dimly fluorescent particles at image edges, while a purely local threshold tends to generate spurious foreground regions in areas of low background variation. The conjunction of both masks keeps only regions that both strategies agree are foreground, which in practice means retaining genuine particles while suppressing most artefacts [21,18].

The selective watershed step addressed particle clustering. Applying watershed indiscriminately to all objects frequently causes over-segmentation of isolated particles. Restricting it to objects whose area exceeds twice the image mean



Fig. 4. MATLAB graphical user interface developed for microplastic analysis, showing both single-image and batch-processing modes.

— a threshold derived from the distribution of the current image rather than a fixed value — reduced both under- and over-counting [24].

It is worth being explicit about what the system does not do. Without spectroscopic information, there is no way to confirm polymer identity from morphology alone [19,2,17]. For applications where polymer type matters, spectroscopic confirmation remains necessary. For screening purposes, however, morphological classification is often sufficient to characterize a sample’s size distribution and shape profile [25,20].

5.2 Batch Processing Results

The batch-processing module was tested on a set of fluorescence microscopy images, processing each through the full pipeline without user intervention. Results for each detected particle were automatically stored in an Excel dataset, while a global HTML report containing statistical summaries and graphical representations was generated at the end of the analysis.

Figure 5 shows the overall class distribution and the morphological size histograms obtained from the batch-processing run. The results reveal wide variability in particle size, with a predominance of small particles, although larger elongated structures were also identified, particularly among particles classified as fibers.

The classification procedure successfully distinguished between fibers, fragments, films, beads and aggregates according to their morphological characteristics. This automatic classification reduces the need for manual inspection and improves processing efficiency when large image datasets are analysed. The implementation of batch processing enabled the automatic analysis of hundreds of images without user intervention, considerably reducing processing time and improving reproducibility compared to manual image analysis approaches.

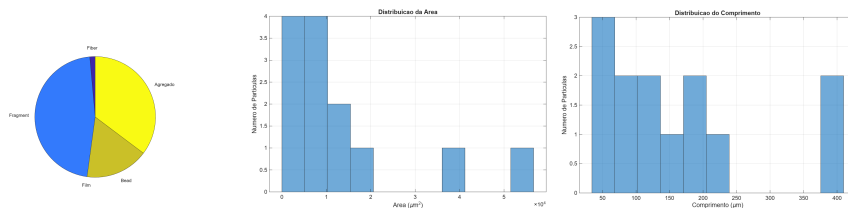


Fig. 5. Batch processing results. Left: global distribution of detected microplastic classes. Centre: distribution of particle areas. Right: distribution of particle lengths.

5.3 Comparison with Related Approaches

The proposed framework deliberately occupies a different point in the accuracy-scalability trade-off space compared to deep learning methods. Approaches such as U-Net or Mask R-CNN achieve higher segmentation precision on their respective training distributions [26,27], but require annotated datasets that are scarce in the microplastic domain [12,6]. The present system requires no labelled data and is deterministic — the same image always produces the same result — which is an advantage in auditable monitoring contexts.

Compared to hybrid spectral-imaging systems [11,10], the proposed pipeline trades chemical identification for simplicity and deployability. For WWTPs where the primary need is particle counting and morphological profiling rather than polymer typing, this trade-off is justified [4,23].

6 Conclusions

The framework presented here achieves what it set out to do: it automates the full microplastic analysis workflow — from raw fluorescence image to classified particle dataset and statistical report — without requiring annotated training data, specialized hardware, or manual intervention at any stage.

The most important methodological contribution is the hybrid thresholding strategy in `bin_itr`, which combines a brightness-adapted Otsu threshold with a local adaptive threshold via logical conjunction. This combination handles the uneven fluorescence distributions typical of real WWTP samples more reliably than either method alone [21,18]. The selective watershed approach further improves particle counts in dense aggregates without introducing over-segmentation artefacts in isolated particles [24].

The rule-based classifier successfully distinguishes five morphological categories using thresholds derived from the physical geometry of each class, rather than from statistical learning. This makes the classification transparent and auditable, which matters in monitoring contexts where results may need to be explained to non-technical stakeholders [12,15].

The batch processing and GUI modules are perhaps less scientifically novel but arguably more important for practical adoption. A method that requires a MATLAB developer to operate is not a monitoring tool — it is a prototype. The

ability to select a folder of images and receive an Excel table and HTML report is what makes the system usable by environmental engineers at a WWTP [23,4].

Looking ahead, the most valuable extension would be validation against spectroscopic ground truth on a representative WWTP dataset, which would allow a quantitative assessment of classification accuracy rather than qualitative visual inspection [12]. Integration with deep learning segmentation for challenging samples, and with spectral imaging for polymer identification, remain natural directions for future work [11,10].

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